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# Improving Emotional Text-to-Speech in Low-Resource Languages via Reinforcement Learning

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Hao-Chun Hsieh Pao-Hua Hsu Nien-Tse Wu

Institute of Artificial Intelligence Innovation

National Yang Ming Chiao Tung University

{handsome.ii14, paohuah.ii14, mattwu.ii14}@nycu.edu.tw

## 1 Project Overview

### 1.1 Profile

**Track: 3. Application**

**Abstract and project goal:** We address the problem of emotional capability loss when an expressive TTS model trained on a high-resource language is adapted to a low-resource language. Although the original model can generate emotionally expressive speech, its emotional controllability and prosodic richness often degrade after fine-tuning on limited target-language data. To tackle this issue, we propose an RL-enhanced adaptation framework built upon IndexTTS2 to preserve and realign emotional expressiveness during low-resource language transfer [Zhou et al., 2025]. Our method aims not to learn emotion from scratch, but to retain the emotional prior of the pretrained expressive model while improving emotion intensity, emphasis control, and speech naturalness in the low-resource language. In our study, we instantiate this setting using Chinese as the high-resource source language and Taiwanese as the low-resource target language.

**TL;DR (“Too Long; Didn’t Read”):** This project studies how to preserve and recover the emotional expressiveness of an expressive TTS model when adapting it from a high-resource language to a low-resource language, using reinforcement learning to improve emotion retention and prosodic control.

### 1.2 Motivation

Expressive speech synthesis is increasingly important for applications such as dubbing, broadcasting, and interactive media. However, when a strong expressive TTS model trained on a high-resource language is adapted to a low-resource language, the model often retains intelligibility but loses emotional expressiveness and fine-grained prosodic control. This makes low-resource adaptation not only a language transfer problem, but also an emotion preservation and alignment problem.

- **Why is the problem interesting?** This problem is interesting because it lies at the intersection of low-resource language adaptation, cross-lingual transfer, and controllable expressive speech synthesis. Rather than learning emotional speech generation from scratch, we focus on preserving and realigning the expressive capability that already exists in the pretrained TTS model when transferring it to a low-resource language.
- **Challenge 1: Emotional Forgetting During Low-Resource Adaptation.** When an emotionally expressive TTS model trained on a high-resource language is fine-tuned on limited low-resource language speech data, it often suffers from catastrophic forgetting, causing the synthesized speech to become flatter and less expressive.

- **Challenge 2: Loss of Fine-Grained Prosodic Control.** After low-resource adaptation, the model may still produce intelligible speech, but fail to preserve natural rhythm, pause structure, emphasis, and emotional intensity that are essential for expressive speech generation.
- **Challenge 3: Difficulty of Cross-Lingual Emotion Alignment.** Emotional cues and prosodic patterns do not transfer perfectly across languages. It is therefore difficult to preserve the expressive prior learned from a high-resource language and align it with the pronunciation, prosody, and linguistic structure of a low-resource language using only limited target-language data.
- **Why the problem remains open or unsolved?** This problem remains open because low-resource languages lack sufficient emotionally annotated speech data for conventional supervised training. As a result, naive fine-tuning often prioritizes pronunciation adaptation over expressive preservation. While seminal foundational models like F5-TTS [Chen et al., 2025] achieve high fidelity, they still lack fine-grained emotional controllability. Conversely, works like EmoSphere++ [Cho et al., 2024] demonstrate strong emotion control but are primarily designed for high-resource scenarios. Extending these capabilities to low-resource languages without catastrophic forgetting remains highly challenging. Several prominent recent works have attempted to enhance expressiveness and language adaptability, including the IndexTTS2 [Zhou et al., 2025] and IndexTTS 2.5 [Li et al., 2026] systems, CosyVoice 3 [Du et al., 2025], and EMORL [Li et al., 2025]. Despite these advances, maintaining deep emotional controllability and prosodic richness during low-resource adaptation without losing the pretrained expressive prior is still an unsolved problem.
- **State-of-the-Art (SOTA) Method:** Currently, IndexTTS2 provides a strong foundation for zero-shot and controllable speech synthesis [Zhou et al., 2025]. In this project, we use it as the expressive base model and investigate how reinforcement learning can help preserve and realign its emotional capabilities during adaptation from a high-resource language to a low-resource language.

## 2 Problem Formulation

We formulate emotional preservation and prosodic realignment during low-resource language adaptation as a fully observable Markov Decision Process (MDP). Operating within an autoregressive Text-to-Semantic (T2S) environment, the RL agent (a language model policy  $\pi_\theta$ ) observes a state  $s$  encompassing available conditioning information: an input text sequence, a speaker timbre embedding, a duration constraint embedding, and a target global emotion embedding. The action  $a$  is the generation of a discrete semantic token sequence, which is subsequently converted into an acoustic waveform. Upon generating a complete sequence, the RL learner receives a multi-objective scalar reward signal  $R(s, a)$  computed from the decoded speech. This composite feedback vector, formalized as  $R = \lambda_1 R_{\text{emo-cls}} + \lambda_2 R_{\text{intensity}} + \lambda_3 R_{\text{emphasis}} + \lambda_4 R_{\text{ASR}}$ , balances four task-specific scalar components: categorical emotion classification accuracy, global emotion intensity (distance to a neutral centroid in Valence-Arousal-Dominance space), local emphasis control (relative pitch and energy patterns), and an ASR-based content reward for semantic intelligibility.

The optimization problem seeks to update the policy parameters  $\theta$  by maximizing the expected multi-objective reward using Group Relative Policy Optimization (GRPO). The maximization objective,  $\mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} [R(s, a) \nabla_\theta \log \pi_\theta(a|s)]$ , is subject to a Kullback-Leibler (KL) divergence penalty constraint,  $\beta \text{KL}(\pi_\theta(\cdot|s) \| p_{\text{adapt}}(\cdot|s))$ . This constraint anchors the policy to a base model  $p_{\text{adapt}}$  to prevent catastrophic deviation and preserve the pretrained high-resource expressive prior. Our formulation rests on two key assumptions: (1) an initial parameter-efficient adaptation stage successfully aligns the model to the target low-resource language while keeping most expressive knowledge intact, and (2) the discrete semantic token space produced by the T2S module is sufficiently expressive to retain and recover subtle emotional intensities and local prosody variations during final acoustic reconstruction.

## 3 Empirical Evaluation

To rigorously evaluate the effectiveness of our proposed RL-enhanced IndexTTS2 framework in preserving and recovering emotional expressiveness during transfer from a high-resource language to a low-resource language, we employ a combination of objective and subjective evaluations.

### 3.1 Performance Metrics

We plan to adopt several key performance metrics to comprehensively assess speech intelligibility, speaker fidelity, emotional expressiveness, and emotion retention after low-resource adaptation.

- **Objective Metrics:**

- **Word Error Rate (WER):** Evaluates speech intelligibility and articulation accuracy using ASR models like FunASR [An et al., 2025] or Whisper [Radford et al., 2022].
- **Speaker Similarity (SS):** Computed as the cosine similarity between speaker embeddings extracted from a pretrained speaker recognition model to ensure the timbre is maintained after low-resource adaptation and RL optimization.
- **Emotion Similarity (ES) / Accuracy:** Measured using the open-source emotion2vec model to objectively verify if the generated speech matches the target emotion category.
- **Emotion Retention Gap (ERG):** Let  $M$  denote an emotion-related metric such as Emotion Similarity or Emotion MOS. We define  $ERG = (M_{\text{base}} - M_{\text{adapt}}) / M_{\text{base}}$  to quantify the relative emotional degradation after low-resource adaptation, and additionally report the recovery ratio  $(M_{\text{ours}} - M_{\text{adapt}}) / (M_{\text{base}} - M_{\text{adapt}})$  to measure how much of the lost emotional capability is restored by our method.

- **Subjective Metrics (MOS Framework):**

- **Multidimensional MOS:** We will conduct human evaluations based on a 1-5 scale for Quality MOS (QMOS), Similarity MOS (SMOS), Prosody MOS (PMOS), and Emotion MOS (EMOS).
- **Emotion Intensity Test (EIT):** A pairwise comparison task where listeners identify the stronger emotional sample among weak, medium, and strong synthesized speech.
- **Emphasis Accuracy Test (EAT):** Assesses the consistency between the model’s predicted emphasis positions (local prosody) and those perceived by human listeners.

The combination of EIT and EAT is crucial as our RL method explicitly optimizes for fine-grained intensity and local emphasis, which standard MOS cannot fully capture.

### 3.2 Baseline Methods

We compare our method against several baselines to explicitly quantify emotional forgetting and emotional recovery during low-resource adaptation.

- **Original High-Resource IndexTTS2 [Zhou et al., 2025]:** The expressive source model before low-resource adaptation, used to establish the upper bound of emotional capability prior to transfer.
- **IndexTTS2 + Naive Low-Resource Fine-Tuning:** A baseline that fine-tunes IndexTTS2 on limited target-language data without any explicit preservation mechanism, used to measure the severity of emotional forgetting.
- **IndexTTS2 + Parameter-Efficient Adaptation:** A target-language adaptation baseline using a lightweight adaptation strategy, used to test whether preserving more pretrained parameters helps retain emotional expressiveness.
- **IndexTTS2 + Parameter-Efficient Adaptation + RL (Ours):** Our full method, which applies RL-based emotional and prosodic alignment after low-resource adaptation.
- **CosyVoice3 [Du et al., 2025]:** A strong zero-shot text-to-speech baseline with predefined emotion instructions for cross-model comparison.

### 3.3 Benchmark Datasets

To train the low-resource adaptation model and evaluate catastrophic forgetting of emotional expressiveness, we employ the following datasets:

- **Emotion Supervision Data:** We utilize the Emotional Speech Database (ESD) and the Espresso dataset, which provide annotations for discrete emotions, continuous intensities, and emphasis. These serve as supervision signals for RL-based emotional realignment training.

- **Taiwanese Across Taiwan Corpus (TAT\_MOE)** [Liao, 2022]: A NAER-hosted corpus from the Ministry of Education project, containing approximately 300k words of native Taiwanese texts (57 authors, about 6,000 characters each) and about 208 hours of parallel recordings (roughly 400 speakers, 4–6 speakers per sentence, recorded with 6 microphones). We use a small subset (1 to 5 hours) of this corpus for low-resource fine-tuning, and additionally draw Taiwanese input texts from the full corpus for evaluation synthesis.
- **Low-Resource Adaptation Task:** The target-language adaptation scenario is instantiated with Chinese as the high-resource source language and Taiwanese as the low-resource target language.

### 3.4 Evaluation Synthesis Protocol

To compare emotional expressiveness across models on Taiwanese, we synthesized a large-scale evaluation corpus from the TAT\_MOE texts (2,752 utterances in Kaldi text format). All systems were conditioned on the same fixed reference speaker audio (FT0BAE.mp3) to control for speaker identity; only the emotion label was varied across the seven target categories: *angry*, *happy*, *fearful*, *disgusted*, *sad*, *surprised*, and *neutral*. Emotion labels were assigned in a balanced, shuffled order across utterances (random seed fixed at 42 for reproducibility). The three systems evaluated are:

- **Taiwanese A2 Model:** A low-resource fine-tuned checkpoint (A2) accessed via a Gradio inference server. Emotion is specified through the natural-language control mode (), passing the emotion label directly as the prompt text. Generation of the full corpus required approximately 7 hours 58 minutes of wall time, producing 7 hours 54 minutes of audio (39 failures).
- **Taiwanese A3 Model:** A subsequent fine-tuned checkpoint (A3) generated under the same setup as A2. Generation required approximately 9 hours 44 minutes of wall time, producing 9 hours 48 minutes of audio (5 failures).
- **IndexTTS2 (Pre-adaptation Baseline):** The expressive Chinese pre-trained model, loaded directly for batch inference. Emotion is specified via an 8-dimensional continuous emotion vector with a fixed intensity of 1.4 (e.g.,  $[0, 1.4, 0, \dots, 0]$  for *angry*). Generation of 2,752 utterances required approximately 1 hour 57 minutes of wall time, producing 9 hours 22 minutes of audio.

In total, the evaluation corpus comprises approximately 27 hours of synthesized Taiwanese emotional speech across all three systems. Emotion recognition accuracy for each system is then assessed using the pre-trained emotion2vec model, which classifies each generated utterance into one of the seven target emotion categories and compares the predicted label against the intended synthesis target.

## 4 Methodology

Our methodology follows a two-stage framework built upon a pre-trained expressive IndexTTS2 model. Starting from a model trained on a high-resource language, we first adapt it to the low-resource target language, and then apply reinforcement learning to preserve and recover emotional expressiveness, prosodic control, and content accuracy during transfer.

### 4.1 Stage 1: Low-Resource Language Adaptation

In the first stage, we adapt the expressive base model to the low-resource target language. The goal of this stage is to acquire target-language pronunciation and linguistic patterns while preserving as much of the pretrained expressive prior as possible. To reduce severe catastrophic forgetting, we adopt a parameter-efficient adaptation strategy, such as lightweight adapter modules or LoRA, so that only a limited subset of model parameters is updated during target-language transfer.

### 4.2 Stage 2: RL-Based Emotional Realignment

In the second stage, we introduce an RL-based alignment procedure [Li et al., 2025] on top of the adapted model. The core idea is to treat the autoregressive generation of semantic speech representations as a sequential decision-making process, while computing task-specific rewards from the decoded speech waveform. This stage explicitly optimizes emotion preservation, emotional intensity, local emphasis, and content intelligibility in the low-resource language. In practice, the RL

updates are mainly applied to the autoregressive T2S transformer, and can optionally include the emotion- or style-related conditioning modules when stronger expressive realignment is needed.

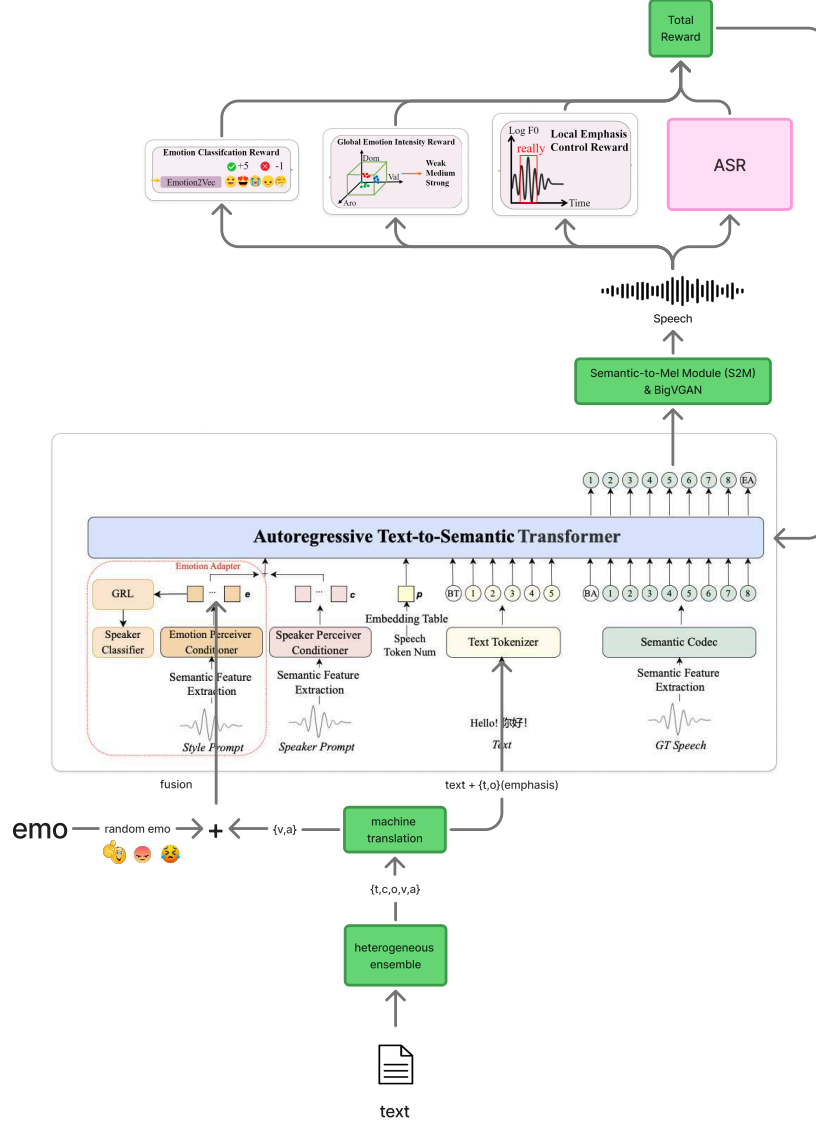


Figure 1: High-level architecture of the proposed RL-enhanced IndexTTS2 framework. After low-resource adaptation, the generator outputs semantic speech tokens, which are decoded into speech waveforms and evaluated by emotion, intensity, prosody, and content reward models. These rewards are used to update the policy via Group Relative Policy Optimization (GRPO), mitigating emotional forgetting during language transfer.

### 4.3 Reinforcement Learning Formulation

The generation of semantic speech tokens is formulated as a Markov Decision Process (MDP), while the reward is computed from the decoded speech waveform:

- **State ( $S_t$ ):** The hidden states of the IndexTTS2 autoregressive decoder, incorporating the input text, speaker conditions, and previously generated speech tokens up to step  $t$ .
- **Action ( $A_t$ ):** The predicted semantic speech token at step  $t$ , sampled from the policy distribution  $\pi_\theta(A_t|S_t)$ .

- **Reward ( $R_t$ ):** A composite reward signal provided upon the completion of a sequence or segment. The reward is designed to preserve emotional expressiveness while maintaining intelligibility in the low-resource target language, and comprises four components:
  1. *Emotion Classification Reward:* Evaluated using a pre-trained emotion recognition model (e.g., emotion2vec) to measure whether the generated speech matches the target emotion category.
  2. *Global Emotion Intensity Reward:* Computed from the distance between the generated speech and a neutral centroid in the Valence-Arousal-Dominance (VAD) space, in order to encourage the target global emotional intensity.
  3. *Local Emphasis Control Reward:* A reward based on the relative pitch and energy patterns of emphasized words, which encourages correct placement of local stress and expressive prosody.
  4. *ASR-Based Content Reward:* Computed using an ASR model on the generated speech to compare the transcription against the input text, encouraging content preservation and intelligibility during RL optimization.

In summary, the sequence-level reward can be written as

$$R = \lambda_1 R_{\text{emo-cls}} + \lambda_2 R_{\text{intensity}} + \lambda_3 R_{\text{emphasis}} + \lambda_4 R_{\text{ASR}},$$

where the coefficients  $\lambda_i$  control the trade-off between emotional expressiveness, local prosody, and content accuracy.

#### 4.4 Alleviating Catastrophic Forgetting

To address the degradation of emotional capabilities during low-resource fine-tuning, we initialize our policy  $\pi_\theta$  with the weights of the target-language adapted IndexTTS2 model rather than training from scratch. We then employ Group Relative Policy Optimization (GRPO) mixed with a Kullback-Leibler (KL) divergence penalty against the adapted reference model. This ensures that the model explores localized emotional and prosodic improvements while preserving the linguistic adaptation and expressive prior inherited from the original high-resource model.

## 5 Preliminary Results

### 5.1 Preliminary Baseline Setup

As an initial pilot study, we conducted a subjective evaluation to establish a preliminary baseline for Taiwanese TTS. The evaluation used the Mean Opinion Score (MOS) on a scale from 1 to 5 (where 5 represents the best performance). The evaluation panel consisted of 12 members from the SARC laboratory. The test corpus involved 20 sentences for evaluating speech intelligibility and naturalness, and an additional 10 sentences specifically for evaluating voice similarity. In total, the subjective evaluation yielded 600 rating samples ((20 \* 2 + 10) test items × 12 evaluators).

### 5.2 Preliminary Baseline Observation

Table 1 shows the preliminary subjective evaluation results of the baseline method, CosyVoice2 [Du et al., 2024], on Taiwanese TTS. The model achieves a similarity MOS of 4.16, an intelligibility MOS of 4.41, and a naturalness MOS of 4.37, resulting in an overall mean MOS of 4.31. These results suggest that a strong off-the-shelf baseline can already generate high-quality Taiwanese speech. This observation motivates our project to move beyond basic intelligibility and naturalness, and to focus on whether emotional expressiveness can be preserved and recovered during low-resource adaptation.

Table 1: Preliminary Subjective Evaluation (MOS) on Taiwanese TTS

| Model      | Similarity MOS ↑ | Intelligibility MOS ↑ | Naturalness MOS ↑ | Mean ↑ |
|------------|------------------|-----------------------|-------------------|--------|
| CosyVoice2 | 4.16             | 4.41                  | 4.37              | 4.31   |

## 6 Expected Contributions of Each Team Member

- **Hao-Chun Hsieh:** Leads the problem formulation and methodology design, establishing the theoretical foundations, and ensuring technical alignment with the overall project objectives.
- **Pao-Hua Hsu:** Directs project documentation and visual communication, including drafting the proposal, developing system architecture diagrams, designing the final poster, and delivering the presentation.
- **Nien-Tse Wu:** Spearheads the empirical evaluation and system implementation, which encompasses dataset curation, experimental setup, and hyperparameter tuning for model optimization.



## References

- Siyi Zhou, Yiquan Zhou, Yi He, Xun Zhou, Jinchao Wang, Wei Deng, and Jingchen Shu. IndexTTS2: A breakthrough in emotionally expressive and duration-controlled auto-regressive zero-shot text-to-speech. *ArXiv*, abs/2506.21619, 2025. URL <https://api.semanticscholar.org/CorpusID:280010717>.
- Yushen Chen, Zhikang Niu, Ziyang Ma, Keqi Deng, Chunhui Wang, JianZhao JianZhao, Kai Yu, and Xie Chen. F5-TTS: A fairytaler that fakes fluent and faithful speech with flow matching. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6255–6271, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. doi: 10.18653/v1/2025.acl-long.313. URL <https://aclanthology.org/2025.acl-long.313/>.
- Deok-Hyeon Cho, H.-S. Oh, Seung-Bin Kim, and Seong-Whan Lee. Emosphere++: Emotion-controllable zero-shot text-to-speech via emotion-adaptive spherical vector. *IEEE Transactions on Affective Computing*, 16:2365–2380, 2024. URL <https://api.semanticscholar.org/CorpusID:273821857>.
- Yunpei Li, Xun Zhou, Jinchao Wang, Lu Wang, Yong Wu, Siyi Zhou, Yiquan Zhou, and Jingchen Shu. IndexTTS 2.5 Technical Report. *arXiv e-prints*, art. arXiv:2601.03888, January 2026. doi: 10.48550/arXiv.2601.03888.
- Zhihao Du, Changfeng Gao, Yuxuan Wang, Fan Yu, Tianyu Zhao, Hao Wang, Xiang Lv, Hui Wang, Chongjia Ni, Xian Shi, Keyu An, Guanrou Yang, Yabin Li, Yanni Chen, Zhifu Gao, Qian Chen, Yue Gu, Mengzhe Chen, Yafeng Chen, Shiliang Zhang, Wen Wang, and Jieping Ye. CosyVoice 3: Towards In-the-wild Speech Generation via Scaling-up and Post-training. *arXiv e-prints*, art. arXiv:2505.17589, May 2025. doi: 10.48550/arXiv.2505.17589.
- Haoxun Li, Yu Liu, Yuqing Sun, Hanlei Shi, Leyuan Qu, and Taihao Li. EMORL-TTS: Reinforcement Learning for Fine-Grained Emotion Control in LLM-based TTS. *arXiv e-prints*, art. arXiv:2510.05758, October 2025. doi: 10.48550/arXiv.2510.05758.
- Keyu An, Yanni Chen, Zhigao Chen, Chong Deng, Zhihao Du, Changfeng Gao, Zhifu Gao, Bo Gong, Xiangang Li, Yabin Li, Ying Liu, Xiang Lv, Yunjie Ji, Yiheng Jiang, Bin Ma, Haoneng Luo, Chongjia Ni, Zexu Pan, Yiping Peng, Zhendong Peng, Peiyao Wang, Hao Wang, Haoxu Wang, Wen Wang, Wupeng Wang, Yuzhong Wu, Biao Tian, Zhentao Tan, Nan Yang, Bin Yuan, Jieping Ye, Jixing Yu, Qinglin Zhang, Kun Zou, Han Zhao, Shengkui Zhao, Jingren Zhou, and Yanqiao Zhu. Fun-ASR Technical Report. *arXiv e-prints*, art. arXiv:2509.12508, September 2025. doi: 10.48550/arXiv.2509.12508.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning*, 2022. URL <https://api.semanticscholar.org/CorpusID:252923993>.
- Y.-F. Liao. Tat\_moe corpus [2022]. Available online at <https://tggl.naer.edu.tw>, 2022.
- Zhihao Du, Yuxuan Wang, Qian Chen, Xian Shi, Xiang Lv, Tianyu Zhao, Zhifu Gao, Yexin Yang, Changfeng Gao, Hui Wang, Fan Yu, Huadai Liu, Zhengyan Sheng, Yue Gu, Chong Deng, Wen Wang, Shiliang Zhang, Zhijie Yan, and Jing-Ru Zhou. Cosyvoice 2: Scalable streaming speech synthesis with large language models. *ArXiv*, abs/2412.10117, 2024. URL <https://api.semanticscholar.org/CorpusID:274762932>.